

Redesigned Order Fulfillment Process

E-Commerce Operations — From 3-Day Delays to Same-Day Dispatch

Problems Being Solved by This Design

P1: 3-Day Average Dispatch Delay | P2: Poor Customer Communication | P3: High Return Rate (Wrong Items)

1. Executive Summary

This document presents the redesigned To-Be order fulfillment process for an e-commerce company experiencing three critical operational failures: an average 3-day delay from order placement to dispatch, a lack of proactive customer communication, and a high return rate driven by incorrect items being shipped.

The redesigned process introduces six automation layers — order intelligence, inventory validation, smart pick-pack, real-time customer communication, last-mile optimisation, and returns management — supported by a technology stack of barcode scanning, OMS/WMS integration, and an event-driven notification engine.

Expected outcomes following full implementation:

- Dispatch delay reduced from 3 days to same-day (< 4 hours from order confirmation)
- Wrong-item rate reduced from ~8% to < 0.5% through mandatory barcode scan validation
- Customer communication touchpoints increased from 1 to 6 per order lifecycle
- Return rate reduced from ~12% to < 4% overall; wrong-item returns to near-zero
- First-attempt delivery success rate improved from 78% to 90%+

2. Root Cause Analysis — What We Are Fixing

2.1 Problem 1: 3-Day Dispatch Delay

- No real-time inventory sync between OMS and WMS — orders queued manually
- Seller acceptance average: 45 minutes; no auto-escalation on non-response
- Pick lists printed at end-of-day batch — morning rush creates 6–8 hr backlogs

- Truck consolidation waits: vehicles depart on schedule, not on volume
- No intelligent routing — all orders treated identically regardless of SLA tier

2.2 Problem 2: Poor Customer Communication

- Single SMS sent only at order confirmation — no updates during fulfillment
- 30% of customer care tickets are pure WISMO ('Where Is My Order?') queries
- No live tracking shared with customer; DE location not visible
- Failed delivery attempts not communicated proactively — customer discovers via app

2.3 Problem 3: High Return Rate (Wrong Items Shipped)

- Manual pick process with paper pick list — no barcode scan verification
- ~8% of orders contain wrong SKU, wrong size, or wrong colour
- Packing station has no system check — QC relies on visual inspection only
- Returns cycle takes 5–7 days; refund takes further 7–10 days post-return receipt

3. To-Be Process Design — Step-by-Step

The redesigned process is organised into 6 phases with 16 numbered steps and 5 decision points (marked ♦). Each step includes the responsible actor, system involved, and output. This section maps directly to the flowchart.

Phase A — Order Intelligence & Automated Confirmation

STE P1	[AUTOMATED] Customer Places Order (Web / App / WhatsApp)  Actor: Customer  System: Storefront + OMS (Order Management System)  Time: Real-time, < 30 seconds  Technology: Smart cart with real-time stock availability check; COD/prepaid auto-routing  Output: Order ID generated; inventory soft-reserved immediately  Improvement vs As-Is: Inventory reserved at cart add — eliminates post-order cancellations from stock mismatch
STE P2	[AUTOMATED] ♦ Payment & Fraud Validation  Actor: OMS + Payment Gateway + Risk Engine  System: Razorpay / PayU + internal risk scoring API

-  **Time:** < 10 seconds
-  **Technology:** Intelligent retry engine with 3-attempt fallback; auto-cancel with instant refund if all fail
-  **If Pass:** Order confirmed; inventory hard-reserved; Seller notified instantly via API push
-  **If Fail:** Customer notified via SMS + app; inventory released; order cancelled
-  **Improvement vs As-Is:** Payment failure loop reduced from avg 12 min to < 1 min; 8% failure rate addressed with smart retry

[AUTOMATED] OMS Generates Optimised Pick Request to WMS

STE
P3

-  **Actor:** OMS → WMS (automated, no human intervention)
-  **System:** OMS + WMS integration via real-time API
-  **Time:** Instant, triggered by payment confirmation event
-  **Technology:** SLA-tiered prioritisation: Express orders → top of pick queue; Standard → batched optimally
-  **Output:** Digital pick request with exact bin location, SKU barcode, quantity, packing instructions
-  **Improvement vs As-Is:** Eliminates end-of-day batch print; pick requests now flow continuously throughout the day

Phase B — Barcode-Validated Pick & Pack

[VALIDATION] Warehouse Associate Receives Digital Pick List (Handheld Scanner)

STE
P4

-  **Actor:** Warehouse Picker + WMS
-  **System:** WMS + Handheld Barcode Scanner (Zebra TC-series or equivalent)
-  **Time:** < 20 min per batch of 10 orders (vs 45–135 min manual)
-  **Technology:** Zone-based routing on scanner — shortest walk path calculated by WMS; velocity-based slotting
-  **Output:** Item physically picked from correct bin
-  **Improvement vs As-Is:** WMS-guided picking replaces paper lists; picker travels optimised path, not random warehouse walk

[VALIDATION] ♦ Mandatory Barcode Scan — Item Verification at Pick

STE
P5

-  **Actor:** Warehouse Picker + WMS
-  **System:** Barcode Scanner → WMS real-time validation
-  **Time:** 3–5 seconds per item
-  **Technology:** Scanner checks: correct SKU, correct size/colour variant, within expiry (if perishable), not recalled
-  **If Match:** Item confirmed; WMS logs pick event; move to packing
-  **If Mismatch:** Scanner alarm; picker directed to correct bin; supervisor alerted; incident logged
-  **Improvement vs As-Is:** CRITICAL FIX for wrong-item problem. Scan validation catches error at source, not at customer doorstep

[VALIDATION] Smart Packing Station — QR Label Print & Pack Verification

STE
P6

-  **Actor:** Packing Associate + WMS
-  **System:** WMS + Label Printer + Packing Station Screen
-  **Time:** 3–8 min per order
-  **Technology:** Station screen shows: order photo, pack dimensions, item list. Second scan at packing confirms match. System prints label only after successful scan
-  **Output:** Sealed package with verified contents + printed shipping label with tracking barcode
-  **Improvement vs As-Is:** Dual-scan verification (pick + pack) creates two independent checkpoints; label only prints if both scans pass

Phase C — Logistics & First-Mile Dispatch

[LOGISTICS] ◆ Carrier Auto-Selection via Logistics Orchestration API

STE
P7

-  **Actor:** WMS → Logistics Orchestration Layer (automated)
-  **System:** Logistics API middleware (Shiprocket / Clickpost / custom)
-  **Time:** < 5 seconds
-  **Technology:** Rules engine selects optimal carrier based on: destination pin code, order weight, SLA tier, carrier performance score, current capacity. Backup carrier pre-assigned automatically
-  **Output:** AWB number generated; tracking link activated; carrier manifest updated
-  **Improvement vs As-Is:** Eliminates manual portal switching across 6+ logistics partners; carrier assignment optimised in real-time

[LOGISTICS] Package Manifested & Dispatched to Pickup Queue

STE
P8

-  **Actor:** Dispatch Team + Logistics Partner
-  **System:** WMS + Carrier API
-  **Time:** Manifest generated in < 2 min; pickup scheduled for next available slot (target: same-day)
-  **Technology:** Dynamic pickup scheduling — carrier pickup triggered by volume threshold, not fixed schedule; GPS confirmation of pickup completion
-  **Output:** Package scanned out; WMS status → 'Dispatched'; tracking activated
-  **Improvement vs As-Is:** Dynamic dispatch replaces fixed-schedule trucks; same-day pickup window guaranteed for Express SLA

Phase D — Proactive Customer Communication Engine

[COMMUNICATION] Event-Driven Notification Engine — 6 Touchpoints

STE
P9

-  **Actor:** Notification Engine (automated) + Customer
-  **System:** Event bus (Kafka/SQS) → Notification Service → SMS / WhatsApp / Email / App Push
-  **Time:** Each notification sent within 60 seconds of triggering event

-  **Technology:** 6 automated touchpoints: (1) Order Confirmed, (2) Packed & Ready, (3) Picked Up by Carrier, (4) Out for Delivery (with live DE GPS link), (5) Delivered + OTP, (6) Feedback Request (24 hrs post-delivery)
-  **Output:** Customer informed at every stage; zero need to contact customer care for status
-  **Improvement vs As-Is:** Increases from 1 to 6 communication touchpoints; WISMO tickets projected to fall by 85%

[COMMUNICATION] Live GPS Tracking — Customer-Facing Delivery Map

STE
P10

-  **Actor:** Delivery Executive (DE) + Customer
-  **System:** DE Mobile App (GPS) → Tracking API → Customer App / Link
-  **Time:** Live, refreshed every 30 seconds from DE app
-  **Technology:** DE app broadcasts GPS co-ordinates; customer sees live map; estimated arrival updates in real-time; 1-tap customer support button on tracking page
-  **Output:** Customer knows exactly where their order is and when it will arrive
-  **Improvement vs As-Is:** Eliminates 'out for delivery but not arrived' complaints; sets precise expectation — reduces failed delivery attempts

Phase E — Smart Last-Mile Delivery

[LOGISTICS] AI-Optimised Route Assignment to Delivery Executive

STE
P11

-  **Actor:** Dispatch Centre + DE
-  **System:** Route Optimisation Engine (Routific / Google OR-Tools / custom)
-  **Time:** Routes computed overnight for next-day batch; updated in real-time for same-day orders
-  **Technology:** AI assigns orders to DEs based on: load weight, geographic cluster, historical delivery performance, time-window preferences selected by customer at checkout
-  **Output:** Optimised route loaded on DE app; customer ETAs pre-calculated
-  **Improvement vs As-Is:** Reduces average route distance by 20–30%; increases orders delivered per DE per day from avg 18 to 25+

[COMMUNICATION] Pre-Delivery Confirmation — 1-Hour WhatsApp Alert

STE
P12

-  **Actor:** Notification Engine + Customer
-  **System:** WhatsApp Business API + DE App location trigger
-  **Time:** Sent when DE is approximately 60 minutes from delivery address
-  **Technology:** Customer receives: ETA, DE name, DE photo, 1-tap reschedule option (if unavailable), OTP pre-alert
-  **Output:** Customer availability confirmed; failed delivery reduced proactively
-  **Improvement vs As-Is:** Pre-delivery alert expected to reduce first-attempt failure rate from 22% to < 10%

STE
P13

[LOGISTICS] ♦ Delivery Attempt — OTP Verification

-  **Actor:** DE + Customer

-  **System:** DE App (offline-capable) + OTP service
-  **Time:** 2–3 min on-doorstep
-  **Technology:** DE app works offline; OTP sent to customer's registered mobile; de scans package barcode at doorstep to confirm correct parcel delivered; delivery photo captured
-  **If Delivered:** OTP entered; WMS → Delivered; settlement trigger sent to Finance; feedback request queued (24 hr delay)
-  **If Failed:** Reason code logged; customer notified; automatic rescheduling offered; max 3 attempts before RTO
-  **Improvement vs As-Is:** Doorstep barcode scan adds final verification checkpoint; OTP replaces manual signature

Phase F — Intelligent Returns Management

[RETURNS] ◆ Return Request — Self-Serve In-App Portal

STE
P14

-  **Actor:** Customer + Returns System
-  **System:** Returns Portal + OMS + Policy Engine
-  **Time:** < 3 minutes for customer to initiate
-  **Technology:** Policy engine auto-checks eligibility (return window, item category, reason code); customer selects reason from structured list; photo upload mandatory for wrong-item/damaged claims
-  **If Eligible:** Return authorised instantly; pickup scheduled within 48 hrs; return label emailed
-  **If Ineligible:** Policy reason shown; escalation path to customer care if disputed
-  **Improvement vs As-Is:** Self-serve replaces call-centre initiation; structured reason codes enable analytics on return root causes

[RETURNS] Doorstep QC by DE at Return Pickup

STE
P15

-  **Actor:** DE + Returns System
-  **System:** DE App + Returns Management System
-  **Time:** 3–5 min at customer door
-  **Technology:** DE app prompts: scan return item barcode, photograph condition, confirm packaging. QC result (Accept / Partial / Reject) recorded in app and synced to WMS
-  **If Accepted:** Refund triggered instantly on scan (processing 2–3 business days to bank)
-  **If Rejected:** Item not accepted; customer informed; dispute process initiated
-  **Improvement vs As-Is:** Refund cycle reduced from 7–10 days to 2–3 days; doorstep QC eliminates disputes about item condition

[INTELLIGENCE] Analytics Loop — Return Reason Aggregation & Root Cause Alert

STE
P16

-  **Actor:** Analytics System → Operations Team
-  **System:** Data Lake + Real-time Dashboard + Alert Engine
-  **Time:** Real-time aggregation; weekly root-cause report auto-generated
-  **Technology:** If wrong-item returns for a specific SKU exceed threshold → automated alert to Warehouse Ops; SKU flagged for manual recount; supplier notified if packaging defect
-  **Output:** Return root causes visible; feedback loop into pick-pack process prevents recurrence

✓ **Improvement vs As-Is:** Closes the loop: returns data now drives proactive process corrections, not just reactive fire-fighting

4. Flowchart Reference Table

Use the table below to draw the To-Be flowchart in Draw.io or Lucidchart. Steps 2, 5, 7, 13, 14 are decision diamonds (◆). All other steps are rounded rectangles. Use yellow for ◆ decision nodes, blue for process steps, and green for terminal end nodes.

Swimlane configuration: Customer | OMS/WMS | Warehouse | Logistics | Communication Engine | Returns

#	Step / Decision	Actor & System	Output / Next Step
1	Customer places order	Customer → Storefront + OMS	Order ID created; inventory soft-reserved
2	◆ Payment & fraud check	OMS + Payment Gateway	Pass → Step 3 Fail → Cancel + refund
3	OMS generates pick request to WMS	OMS → WMS (automated)	Digital pick request with bin location sent to WMS
4	Picker receives digital pick list on scanner	Warehouse + WMS	Picker routes to optimal bin
5	◆ Barcode scan — item match check	WMS + Barcode Scanner	Match → Step 6 Mismatch → Alert + redirect
6	Smart packing station — second scan + label print	Packing Associate + WMS	Package sealed; label printed; ready for dispatch
7	◆ Carrier auto-selection via logistics API	Logistics API	AWB generated; tracking link activated
8	Manifest & dispatch to carrier pickup	Dispatch + Carrier	Package scanned out; WMS status → Dispatched
9	Notification: Order Confirmed + Packed + Dispatched	Notification Engine	3 automated messages sent to customer

10	Live GPS tracking link sent to customer	DE App + Tracking API	Customer can view live DE location
11	AI route assignment to DE	Route Optimiser + DE App	Optimised route loaded; ETAs set
12	Pre-delivery WhatsApp alert (60 min ETA)	Notification Engine + DE App	Customer confirms availability / reschedules
13	◆ Delivery attempt — OTP capture	DE + OTP Service + DE App	Delivered → Step 16 Failed → Reschedule (max 3x) RTO after 3 fails
14	◆ Customer initiates return request	Customer + Returns Portal	Eligible → Pickup scheduled Ineligible → Care escalation
15	DE doorstep QC at return pickup	DE + Returns App	Accepted → Refund triggered Rejected → Dispute
16	Analytics: return reason logged + root cause alert	Analytics Engine → Ops Team	Feedback loop to warehouse ops; SKU alerts

5. Recommended Technology Solutions

Layer	Tool / Solution	What It Solves
Order Management	Custom OMS or Unicommerce / SellerApp	Real-time inventory reservation; multi-channel order sync; SLA tiering
Warehouse Mgmt (WMS)	SnapFulfil / Increff / Manhattan WMS	Digital pick lists; velocity slotting; bin management; dual-scan validation
Barcode Scanning	Zebra TC52 / Honeywell CT40 handhelds	Item verification at pick and pack — eliminates wrong-item dispatch
Logistics Orchestration	Shiprocket / Clickpost / custom API layer	Single API for all carriers; auto-selection; AWB + tracking integration
Notification Engine	Exotel / Gupshup + WhatsApp Business API	Event-driven 6-touchpoint customer communication; WISMO elimination

Customer Tracking	Custom tracking page + DE GPS broadcast	Live DE location; ETA updates; 1-tap support
Route Optimisation	Routific / LogiNext / Google OR-Tools	AI-optimised DE routes; pre-delivery ETA; load balancing
Returns Platform	Returnly / custom returns portal	Self-serve returns; doorstep QC; instant refund trigger
Analytics & Alerting	Metabase / Apache Superset + Kafka streams	Real-time ops dashboard; return root cause alerts; SLA breach prediction
DE Mobile App	Custom (offline-capable, React Native)	Offline scan + OTP + photo; GPS broadcast; return QC workflow

6. Expected KPI Improvements

The table below shows projected performance improvements following full implementation. Quick Win improvements are expected within 90 days; full targets by Month 12.

KPI Metric	As-Is (Current)	To-Be (Target)	Improvement
Average dispatch time	3 days	< 4 hours	95% reduction
Order-to-doorstep (metro)	5–7 days	1–2 days	70% faster
Wrong-item rate	~8%	< 0.5%	94% reduction
Overall return rate	~12%	< 4%	67% reduction
WISMO customer care tickets	~30% vol	< 5% vol	83% reduction
First-attempt delivery success	~78%	> 90%	+12 pp
Customer notification touchpoints	1 per order	6 per order	6× increase
Return pickup SLA	5–7 days	< 48 hours	75% faster
Refund cycle time	7–10 days	2–3 days	70% faster

Seller settlement	T+11	T+2	80% faster
Pick accuracy rate	~92%	> 99.5%	+7.5 pp
DE orders per day (avg)	~18	25+	+39%

7. Implementation Roadmap

The roadmap is structured in three phases. Phase 1 (Quick Wins) delivers the highest-impact changes within 90 days using minimal technology investment. Phase 2 (Core Build) implements the technology foundation. Phase 3 (Intelligence) adds the analytics and optimisation layer.

Phase	Initiative	Technology	Timeline	Expected Gain
Quick Win	Implement barcode scanning at pick station	Zebra/Honeywell handhelds + WMS scan module	Days 0–30	Wrong-item rate → < 0.5%
Quick Win	Deploy omnichannel notification engine	WhatsApp Business API + SMS (Exotel/Gupshup)	Days 0–45	WISMO tickets → < 5%
Quick Win	Seller SLA enforcement + auto-escalation	OMS config change + seller dashboard alert	Days 0–30	Acceptance time → < 15 min
Quick Win	Self-serve return portal (basic)	Simple web form + policy rules engine	Days 30–60	Return initiation < 3 min
Quick Win	Dynamic pick list (continuous vs batch)	WMS config + queue rules	Days 0–20	Pick backlog eliminated
Core Build	OMS–WMS real-time API integration	Custom integration layer / iPaaS (MuleSoft/Boomi)	Months 2–5	Dispatch time → < 4 hrs
Core Build	Logistics orchestration API layer	Clickpost / Shiprocket multi-carrier API	Months 2–4	Carrier selection time → 5 sec
Core Build	DE offline-capable mobile app (v2)	React Native + offline sync + GPS broadcast	Months 3–6	DE uptime in low-connectivity zones

Core Build	Live customer tracking page + GPS	Tracking API + customer web view	Months 3–5	Delivery anxiety eliminated
Core Build	Smart packing station (dual scan + screen)	Packing workstation upgrade + WMS label print	Months 2–3	Second verification checkpoint
Intelligence	AI route optimisation engine	Routific / Google OR-Tools API	Months 6–9	+39% orders per DE per day
Intelligence	Real-time ops dashboard + SLA alerting	Metabase / Superset + Kafka event streaming	Months 6–10	Proactive delay intervention
Intelligence	Return analytics + root cause alert engine	Data lake + anomaly detection model	Months 8–12	Return prevention vs correction
Intelligence	Demand forecasting + inventory pre-positioning	ML model on SKU velocity data	Months 9–12	Stock-outs reduced 60%

8. Risk & Mitigation Plan

Risk	Likelihood	Impact	Owner	Mitigation
Scanner adoption resistance by warehouse staff	High	Medium	Ops Manager	Gamify pick accuracy (leaderboard + bonus); run 2-week parallel (paper + scan); floor trainer on each shift for first month
WMS integration with legacy OMS takes longer than planned	Medium	High	CTO	Begin integration in Month 2 with iPaaS middleware; fallback: file-based daily sync as bridge during transition
WhatsApp Business API approval delays (Meta)	Medium	Medium	Tech Lead	Begin application Week 1; use SMS-only as fallback; switch to WhatsApp when approved
Return portal misuse / fraudulent returns	Medium	High	Finance + Ops	Mandatory photo upload for all returns; ML fraud scoring on

				return patterns; doorstep QC rejects damaged/used items
Carrier API instability during peak sale events	Low	High	Logistics Head	Pre-assign backup carrier for every order; circuit breaker in API layer; manual fallback queue with 15-min SLA
DE app battery / connectivity issues in Tier 2/3	High	Medium	Field Ops	Build full offline capability (OTP, scan, photo work without internet); sync on next connectivity window
Customer confusion with OTP delivery process	Medium	Low	Customer Care	Clear communication campaign; OTP instructions on tracking page and notification; grace period with signature fallback
Data quality issues in existing inventory records	High	High	Warehouse Mgr	Mandatory full inventory audit (cycle count) before WMS go-live; freeze changes during count window

9. Change Management & Quick Wins Timeline

Week 1–2: Foundation

- Kick off barcode scanner procurement (Zebra TC52 × 20 units, est. cost: ₹8–12 L)
- Apply for WhatsApp Business API — begin Exotel SMS notification engine setup
- Assign workstream owners: Warehouse Ops, Logistics, Tech, Customer Experience
- Run full warehouse cycle count (mandatory prerequisite for WMS accuracy)

Week 3–4: First Wins Deployed

- Go live: barcode scanning at pick station (wrong-item fix)
- Go live: dynamic pick list queue (eliminates morning batch backlog)
- Go live: SMS notification engine — order confirmed, packed, dispatched
- Seller SLA dashboard + auto-escalation rules activated

Month 2–3: Communication & Returns

- Go live: self-serve return portal (basic eligibility check + pickup scheduling)

- Go live: WhatsApp Business API notifications (6 touchpoints)
- Smart packing station installation and training
- Begin OMS-WMS integration development (estimated 10-week sprint)

Month 4–6: Core Technology

- OMS-WMS real-time API integration go-live
- Logistics orchestration API: all carriers on single integration
- DE app v2 go-live with offline capability and GPS broadcast
- Live customer tracking page launched

Month 7–12: Intelligence Layer

- AI route optimisation deployed (start with top-5 metro cities)
- Real-time ops dashboard + SLA breach alerting
- Return analytics engine + root cause alert loop
- Demand forecasting pilot (top 500 SKUs)

10. Success Metrics & Review Cadence

30-Day Checkpoint (Quick Wins)

- Wrong-item rate: was 8%, target < 2% (barcode scanning live)
- WISMO tickets: was 30%, target < 15% (basic SMS notifications live)
- Pick backlog cleared by 9 AM daily (dynamic queue live)

90-Day Checkpoint

- WISMO tickets: target < 5% (WhatsApp 6-touchpoint live)
- Return rate: target < 8% (wrong-item returns near-zero)
- Average dispatch time: target < 8 hours (dynamic pick + smart packing)

12-Month Target State

- Dispatch time: < 4 hours from order confirmation
- Wrong-item rate: < 0.5% — near-zero
- Overall return rate: < 4%
- First-attempt delivery success: > 90%
- WISMO tickets: < 5% of total care volume

- Customer NPS improvement: +15–20 points (driven by communication + accuracy)

Review Cadence

- Weekly: Dispatch SLA, wrong-item rate, WISMO volume (Operations review)
- Bi-weekly: KPI dashboard review with CTO + Ops + Logistics heads
- Monthly: Full fulfillment performance review + roadmap milestone check
- Quarterly: Customer NPS survey + Return root cause analysis

— End of To-Be Process Design Report —